

Grid of radiative transfer models: dense cores embedded in clouds

Take the dust temperature T and the attenuating column Σ_a of every shell

Fit $T(\Sigma_a) = T_f + (T_0 - T_f)(1 + \Sigma_a/\Sigma_\star)^{-\gamma}$, Eq. (7)

Trial line of sight of surface density Σ , one uniform structure

Attenuating column $\Sigma_a(u)$ at every depth, averaged over directions, Eq. (6)

Temperature $T(\Sigma_a(u))$ at every depth, from the fitted relation

Emission of all the depths added, giving the intensities $I_\lambda(\Sigma)$, Eq. (8)

One modified blackbody fitted as the image was, giving Σ_f , Eq. (9)

Tabulate $f(\Sigma_f) = \Sigma/\Sigma_f$, what an image holds, Eq. (10)

Interpolate f onto the image $\mathcal{D}$ at the coarsest resolution

Add $(f-1)\mathcal{D}$ to that image and to every finer one, Eq. (12)